Raj Kamal Mehta

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EXPERIENCE

AMSN Syntax IT Solution Pvt. Ltd. |*Mobile & Web Development Intern* Jan 2026 - Apr 2026

- **Tech stacks used:** React Native, React.js, REST APIs
- **Frontend Architecture:** Engineered responsive web and mobile screen layouts using React.js and React Native, translating design requirements into highly intuitive and user -friendly interfaces.
- **UI/UX Optimization:** Diagnosed and resolved critical UI/UX bugs across existing web applications, improving cross-browser compatibility and overall platform stability.

IIIT Naya Raipur |*Machine Learning (ML) & Deep Learning Trainee* July 2025 - Aug 2025

- **Tech stacks used:** Python, React Native, FastAPI, Kotlin, Scikit-Learn
- **Full-Stack Architecture:** Architected a cardiovascular health application extracting raw PPG optical signals directly from a smartphone camera using custom Kotlin Native hardware bridges.
- **Clinical ML Pipeline:** Engineered a Machine Learning pipeline applying Butterworth DSP filters and extracting 30+ statistical features to achieve 95% classification accuracy with a Random Forest model deployed on FastAPI.

PROJECTS

MittiCard - Soil Health App |*React Native, PostgreSQL, Groq Vision* Feb 2026 - Apr 2026

- Developed an AI-powered mobile app that scans ICAR Soil Health Cards using Groq Vision OCR and delivers crop-specific fertilizer recommendations with cost estimates in Hindi and English.
- Engineered an advisory system for 7 crops with NPK deficiency detection and soil health scoring (0-100), supported by a Node.js/Express and PostgreSQL backend with OTP authentication and Firebase push notifications.

Bike Rental App - Booking Platform |*React Native, Node.js, MongoDB, Stripe* Dec 2025 - Jan 2026

- Architected a cross-platform mobile application supporting end-to-end booking workflows, secure Stripe payments, and automated KYC via REST APIs.
- Designed a scalable Node.js backend with JWT authentication and optimized MongoDB schemas, reducing query response time by 30% under concurrent load.

PPG Signal Quality Assessment |*React Native, FastAPI, Scikit-Learn, Kotlin* July 2025 - Aug 2025

- Architected a full-stack cardiovascular health application that extracts raw PPG optical signals directly from a smartphone camera using custom Kotlin Native hardware bridges.
- Engineered a clinical Machine Learning pipeline applying Butterworth DSP filters and extracting 30+ statistical features to achieve 95% classification accuracy with a Random Forest model deployed on FastAPI.

EDUCATION

Krishna Vikas Institute of Technology , Raipur 2022 - 2026
B.Tech, Computer Science and Engineering *CGPA: 8.3/10*

SKILLS

Languages : JavaScript, TypeScript, Python, C
Frontend : React.js, Redux, HTML5, CSS3, Tailwind CSS
Backend : Node.js, Express.js, REST APIs, JWT Authentication
Mobile Dev : React Native (CLI, Expo)
Databases : MongoDB, PostgreSQL, MySQL
Tools : Git, GitHub, Docker, CI/CD, Postman, Firebase, VS Code

ACHIEVEMENTS

- **Publication:** Raj Kamal Mehta et al., "PPG Signal Quality Assessment & Artifact Detection for Smartphone Monitoring," TechRxiv (IEEE-supported preprint), 2025.
- **Certificate:** Artificial Intelligence & Machine Learning Training, Infosys Springboard (2024)
- **Certificate:** Job-Ready MERN Full-Stack Web Development, 30 Days Coding (Nov 2024)
- **Hackathon Finalist:** IDEA-FIESTA Hackathon (2022)